

SA5 KSM and Cloud: Twelve Complete Contribution Drafts

完整技术草稿合订本 / *Internal drafting edition* / 2026-09-01

COMPLETE TECHNICAL DRAFT · INTERNAL REVIEW

本册展开上一版选定的 12 篇优先提案，不是把 36 个备选 solution 各拆成独立 TDoc。每篇均有独立理由、拟插入 TR 的完整 solution、信息语义、procedure、异常处理、评价与请求结论。正文是英文，可拆分到正式会议模板。图中角色是逻辑职责，不预设新增标准实体。

共同编辑说明：正式 TDoc 文号、下一次会议的日期地点、Source 和议程编号须由提交单位注册填写，本文不冒用公司或会议身份。Target 为 TR 32.801-01；依据 SA5#168 本地定稿，而非声称已核对完整 v0.4.0 编辑稿。各篇标记“Proposed text”之后是候选研究文本，不是现行规范。Solution 编号由 editor 分配；本稿用内部 T01 至 T12 定位，不制造正式 8.x 编号。技术文本已写全，缺失的仅是行政元数据和投稿前的编辑基线定位。

投稿前须检查同类 solution 是否已出现、既有 MnS 是否覆盖新增语义，并由本方完成 IPR 及专利公开审查。本册没有声称方案已经实验验证或取得会议共识。

CONTENTS

- T01 Context-aware contribution and commit for shared knowledge
- T02 Validation feedback and publication gating for knowledge contributions
- T03 Impact-based knowledge revalidation and consumer feedback
- T04 Contextual conflict detection and unresolved-result handling
- T05 Cause-aware knowledge lifecycle transitions
- T06 Revision-consistent knowledge query via context
- T07 Knowledge health evidence and purpose-specific assessment
- T08 Mapping provisioning requests to declarative deployment outcomes
- T09 Partial failure and safe compensation in declarative LCM
- T10 Temporal association of NF, deployment and runtime observability
- T11 Resource-bounded observability control with applied-coverage feedback
- T12 Sub-network LCM coordination using external NF deployment management
- Annex A. Submission checks and source register

T01 Context-aware contribution and commit for shared knowledge

Document status: complete technical draft for discussion and agreement. Target: TR 32.801-01, Knowledge Sharing, with dependencies on knowledge architecture, validation and conflict handling. Source and official TDoc number: to be assigned by the submitting organization.

1. Reason for contribution

S5-263546 retains the objective that multiple knowledge producers can contribute to and consume a common knowledge instance, and that one producer becoming unavailable should not prevent the others from accessing that instance. The detailed sharing-group procedure in the earlier KSM1 material did not survive in S5-263545. Consequently, neither a mandatory knowledge-authority role nor the earlier group-membership procedure is assumed here.

Concurrent contributions need more than a comparison of modified fields. Two contributions that change different assertions may jointly violate a capacity, exclusivity or applicability constraint. Conversely, rejection of every contribution based on an older revision unnecessarily serializes unrelated work. This contribution defines an externally observable submission, validation and publication boundary, leaving storage, replica coordination and constraint-evaluation algorithms implementation specific.

2. Proposed text: context-aware shared-knowledge contribution

2.1 Scope and principles

This solution supports contribution by two or more participants to a common knowledge instance. A participant may consume the published result as well as contribute content. The receiving knowledge producer exposes the contribution capability and is responsible for identifying the published result. A validation capability may be internal or delegated; delegation does not transfer the receiving producer's responsibility for the result it exposes.

The receiving producer is a logical service boundary and may have redundant instances. This solution does not require a new architectural entity called a knowledge authority. A contribution becomes visible as part of the common instance only after its resulting candidate view has passed the applicable validation. Receipt, validation and publication are distinguishable outcomes.

2.2 Information associated with a contribution

Information	Meaning and handling
Contribution correlation identifier	Identifies one logical submission within the authorized participant scope. Reuse with different content is rejected.
Knowledge instance reference and base revision	Identifies the target and the version on which the participant based its modification.
Modification and applicability scope	Identifies proposed additions, replacements or withdrawals and their entity, time and domain context.
Preconditions and semantic profile	Identifies assumptions, applicable vocabulary/mapping version and constraints that the receiving producer can evaluate.
Partial-acceptance preference	States whether an independently valid subset may be accepted. Absence means that the submission is evaluated as one unit.

Result reference	Allows retrieval of acceptance, validation and publication outcomes without requiring a mandatory KnowledgeJob object.
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The semantic names above describe information requirements, not a decided encoding or new mandatory NRM attributes. Authorization is checked before disclosing target revisions or conflict details. Conflict evidence returned to a contributor is restricted to content the contributor is entitled to see.

2.3 Procedure

- Step 1. Participant A retrieves a published knowledge revision r and its semantic context, or otherwise obtains an authorized reference to that revision.
- Step 2. A submits a contribution identifying r, the intended modification, its context, preconditions and a correlation identifier. The receiving producer checks identity, authorization, supported profile and request consistency. A duplicate request with the same identifier and content returns the existing result.
- Step 3. The producer records the request as received and constructs a candidate view. The candidate is not exposed as valid shared knowledge. An asynchronous response identifies where the participant can retrieve the final result.
- Step 4. The producer checks intervening changes since r. It considers both direct overlap and constraints that depend on the changed content. If dependency coverage cannot be established, it expands the validation scope or requires a refreshed submission instead of assuming independence.
- Step 5. The validation capability checks the candidate view under the applicable semantic profile, context and quality acceptance conditions. Different contexts may permit apparently contradictory assertions to coexist. A genuine unresolved conflict produces an unsuccessful or pending-resolution result, not an arbitrary winner.
- Step 6. Before publication, the producer checks that the revision and conditions used for validation are still applicable. If they changed, it repeats the affected checks or returns a retry result. Otherwise, it publishes one identifiable new revision and records the relationship to the contribution result.
- Step 7. The producer returns the published revision, accepted scope and validation reference to A. If partial acceptance was permitted, the result identifies accepted and unaccepted portions and demonstrates that the accepted portion is independently valid.
- Step 8. Other authorized participants retrieve the committed revision. Loss of A does not remove that revision or make A the sole access path.

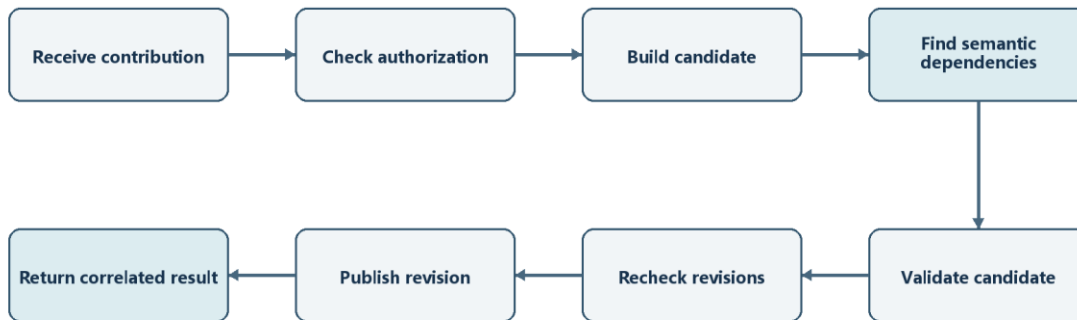
T01 - Safeguarded processing flow

Figure T01-1. Logical interaction and safeguarded processing flow.

2.4 Failure, retry and alternative coordination

Loss of a response is resolved by looking up the contribution identifier; it does not authorize an unqualified second publication. If the receiving producer fails after publication but before replying, its recovered or redundant instance returns the durable publication result. If that result cannot be established, the operation remains indeterminate until reconciled.

A network partition may leave already committed, still-valid revisions readable while preventing new writes. This solution does not promise write availability in every partition. Access to an older committed revision remains subject to current validity and authorization controls.

As an alternative deployment, independent context partitions may be written concurrently. A contribution crossing a shared constraint needs coordinated candidate validation; different fields or partitions alone do not prove independence. A federated participant may also submit a proposal for later acceptance into the common instance. Such a proposal remains distinct from published shared knowledge. These alternatives preserve the same result semantics and do not redefine the producer/consumer roles.

2.5 Example and evaluation

Producer A proposes switching a cell to an energy-saving mode. Producer B changes the minimum capacity required for a neighbouring service. Their modified assertions differ, but both affect a coverage/capacity constraint for the same time interval. The receiving producer evaluates the combined candidate before publication. If the interval differs, both assertions may be retained with distinct applicability contexts; otherwise, the unresolved combination is returned for correction.

Evaluation should compare full serialization, field-only conflict detection and context-aware validation using identical contribution sequences. Relevant criteria are undetected semantic conflicts, unnecessary rejection, validation work, publication latency and recovery after response loss. No numerical gain is claimed by this contribution.

3. Impact, references and proposed agreement

Potential MnS impact is limited to contribution correlation, base-revision/precondition semantics and distinguishable validation/publication results where existing capabilities do not provide them. No new sharing-

group IOC, query language, mandatory database or replication protocol is proposed. Relevant sources are S5-263545, S5-263546 and S5-263547 as finalized at SA5#168; HTTP conditional requests and Kubernetes field management are implementation background, not proposed mandatory dependencies.

SA5 is invited to agree to include the proposed solution as a candidate for Knowledge Sharing and to evaluate its information requirements together with the validation and conflict-handling solutions. Assignment of a final solution number and any mapping to existing operations is an editorial and follow-on study task.

T02 Validation feedback and publication gating for knowledge contributions

Document status: complete technical draft. Target: TR 32.801-01, Knowledge Validation. Related issues: architecture, sharing and health information. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263546 includes validation before knowledge creation and during its lifecycle, exposure of valid knowledge with associated validation information, and validation feedback following a contribution. The meeting record includes a question about who performs validation. A received contribution, a successful validation and an exposed knowledge revision are different events. Without an explicit boundary, a contributor or consumer can mistake asynchronous acceptance for evidence of validity.

This contribution defines the responsibility and procedure for validation at the contribution entry point. It applies whether the contributor is called a producer or is acting as a consumer of the contribution capability. It does not require a centralized validator for all management domains.

2. Proposed text: contribution validation and publication gating

2.1 Responsibility and applicability

The knowledge producer exposing a revision is responsible for applying a publication decision to that revision. It may execute validation internally or use a delegated validation capability. A delegated result identifies what was validated, against which profile and input versions, and at what time. The exposing producer determines whether the result is still applicable when publication occurs.

A contributor's self-validation may be used as evidence but does not automatically authorize exposure. The producer may accept that evidence under a defined trust and authorization policy; the policy and evidence applicability are checked in the same manner as other validation inputs. A health characteristic is an input to a relevant acceptance condition and is not, by itself, a universal validity decision.

2.2 Logical processing outcomes

The procedure distinguishes received, validation in progress, validated, rejected, and published outcomes. These labels describe operation progress, not additional lifecycle states of a knowledge instance. A received or validated candidate is not available through ordinary retrieval as published valid knowledge until the publication gate succeeds.

Information	Required semantics
Candidate reference	Identifies the exact content or immutable candidate revision under validation.
Validation profile and context	Identifies the schema/semantic rules, applicability and relevant acceptance conditions.
Validation result	Indicates pass, fail or indeterminate, with evaluated scope and reason categories.
Evidence versions	Identifies the source, mapping and constraint versions needed to establish applicability of the result.
Publication outcome	Identifies the exposed revision, or explains why validation did not lead to publication.

An indeterminate result covers unavailable evidence or unsupported checks. It is not silently converted into pass. Sensitive validation evidence can be withheld while a non-sensitive reason category is returned.

2.3 Procedure

- Step 1. The contributor sends candidate knowledge or a proposed modification, with a correlation identifier, target/base reference where applicable, and the context needed for evaluation.
- Step 2. The receiving producer checks authorization, input completeness and supported semantics. It rejects malformed or unauthorized requests before invoking expensive validation. For a well-formed request, it creates an immutable candidate reference or an equivalent stable representation.
- Step 3. The producer determines applicable checks. These may include structure, permitted vocabulary, relation consistency, context compatibility and purpose-relevant quality acceptance. A check that is not applicable is distinguished from a check that could not be executed.
- Step 4. The producer invokes its validation capability with the candidate and a stable set of relevant versions. If validation is delegated, the request binds the delegate's result to those inputs and includes a response deadline where appropriate.
- Step 5. The validation capability returns the outcome, affected scope, check results and evidence applicability. A callback referring to another candidate or an expired validation attempt is not used to authorize the current candidate.
- Step 6. The producer applies the publication gate. It confirms that permissions, current target revision, semantic profile and relevant evidence have not invalidated the result. A change causes revalidation of the affected checks or a failed/indeterminate operation.
- Step 7. On success, the producer publishes the resulting revision with its validation reference. On failure, it retains the candidate only as required by authorized retention and diagnostic policy, or deletes it; the candidate is not exposed as valid knowledge.
- Step 8. The producer provides final feedback to the contributor, identifying whether the candidate was rejected, validated but not published, or published. Retrieval of an asynchronous result is possible independently of whether KnowledgeJob is used as an implementation representation.

T02 - Safeguarded processing flow

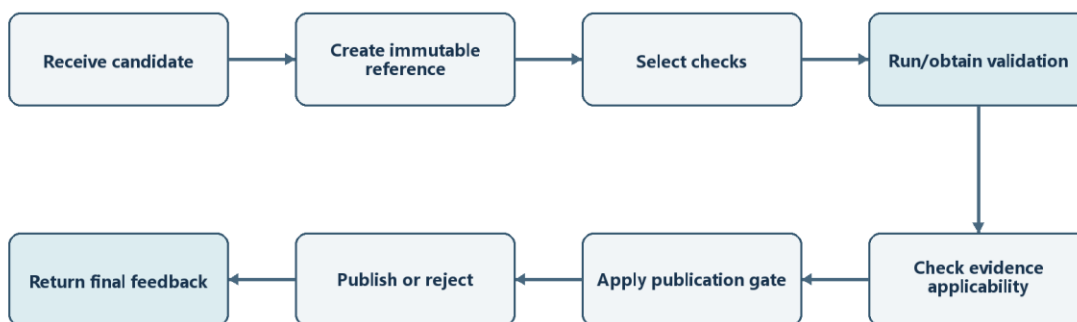


Figure T02-1. Logical interaction and safeguarded processing flow.

2.4 Exceptions and partial processing

If a validator times out, the producer reports indeterminate or continues processing within the agreed deadline. It does not reuse a validation result from different content. If the contributor retries, the correlation identifier resolves to the original operation unless the request explicitly starts a new candidate.

Partial acceptance is permitted only when authorized by the contributor and when the accepted portion can be independently validated. A schema-valid fragment that depends on a rejected semantic assertion is not independently valid. If the target cannot represent separately valid portions, the whole candidate is processed as a unit.

If publication fails after validation, the producer returns a distinct non-publication result. A retry may reuse the validation result only after verifying that its inputs and context remain applicable. A result-delivery failure after publication is recovered by operation lookup, not by publishing another revision.

2.5 Example and evaluation

A contributor submits a relationship between an alarm pattern and a remediation action. The syntax is valid, but the referenced NF has changed its operating mode during validation. The producer rechecks the relevant mode-dependent constraint before publication. If evidence is unavailable, the candidate remains unexposed and the contributor receives an indeterminate result. This prevents a technically well-formed but contextually unsupported recommendation from entering a closed loop.

Evaluation covers source changes during validation, delegated responses arriving out of order, partial acceptance with hidden dependencies, and failure between validation and publication. Criteria include incorrect publication, unobservable validation failure, contributor recovery and avoidable repeated validation. Implementation efficiency is evaluated separately from the semantics of the gate.

3. Impact, references and proposed agreement

The proposal clarifies producer responsibility and operation results. It does not mandate a new validator MnF, a new standalone validation service, SHACL, a universal health score or additional knowledge lifecycle states. Existing capabilities for asynchronous results and notifications should be reused where their semantics suffice. Sources: S5-263546 and S5-263547; W3C SHACL is an example of existing constraint-validation technology, not the required realization.

SA5 is invited to include this candidate solution under Knowledge Validation and to use the separation of receipt, validation and publication when evaluating related sharing and generation solutions.

T03 Impact-based knowledge revalidation and consumer feedback

Document status: complete technical draft. Target: TR 32.801-01, Knowledge Validation; related issue: Knowledge Lifecycle Management. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263546 retains notification to previous consumers when consumed knowledge becomes suspected invalid. It also retains generation provenance information, although the independent Knowledge Lineage KI was removed. A useful procedure should therefore identify enough dependency information to limit invalidation, without requiring a complete historical lineage service. A consumer may have read only part of an instance or used it for a particular action, so the producer's ability to establish an affected boundary is important.

This solution addresses changes after publication. It complements entry validation and does not turn a consumer's use of old knowledge into a permanent entitlement to continue using it.

2. Proposed text: impact-based revalidation

2.1 Information and responsibilities

The exposing producer maintains or can obtain dependency evidence sufficient for its chosen validation granularity. A dependency may identify a source revision, semantic mapping, constraint, context or time condition. It need not retain all source data or every intermediate reasoning step.

Where supported and authorized, a knowledge provision result is associated with a consumption reference that binds the revision and provided scope. A purpose or intended action category may be included where needed to assess applicability. The consumer retains its local relationship between consumed knowledge and management decisions; detailed business logic need not be disclosed to the producer.

Information	Meaning
Trigger and affected evidence	Identifies the changed source, rule, profile, feedback or validity condition.
Knowledge impact scope	Identifies the revisions and independently assessable portions that may no longer be valid.
Impact event identifier	Supports duplicate detection and reconciliation of notifications and query results.
Consumption reference	Identifies previously provided content without requiring disclosure of unrelated consumer activity.
Revalidation result	Identifies restored validity, replacement revision, persistent invalidity or unresolved uncertainty.

2.2 Procedure

- Step 1. The producer publishes validated knowledge and, when providing it, records the minimum authorized information needed to identify previous consumers and provided scope. If consumption tracking is unavailable, the limitation and supported recovery mechanism are made explicit.
- Step 2. A source revision is withdrawn, a constraint changes, a time condition expires, or relevant consumer feedback is received. The producer authenticates and classifies the trigger. Untrusted feedback may start an investigation without being treated as proof of invalidity.

- Step 3. The producer identifies potentially affected knowledge from dependency evidence. If it cannot prove a narrower boundary, the impact scope is expanded to the instance or other safely assessable unit.
- Step 4. The producer marks the affected scope suspected invalid and prevents ordinary exposure of that scope. If the model cannot represent independent validity of portions, the entire instance is suspended as applicable. A known-invalid portion is not kept accessible merely to preserve service availability.
- Step 5. The producer sends an impact notification to identifiable previous consumers whose provision scope intersects the impact. The notification includes the impact event, affected revision/scope, reason category and available recovery reference. It is not a newly assumed independent Health-change notification.
- Step 6. Each consumer reconciles the notification with local knowledge usage. It pauses unexecuted actions that require the affected knowledge, obtains new evidence or requests revalidation. For an already executed action, the consumer evaluates an authorized corrective action; the notification does not automatically undo physical network changes.
- Step 7. The producer revalidates the affected scope against the current evidence. If validated, it publishes or identifies the applicable revision. Otherwise, it maintains the exposure restriction and provides the failed or indeterminate result.
- Step 8. Consumers retrieve the result, update their local decision dependencies and resume only actions whose conditions are satisfied. Duplicate and late events are reconciled by their affected revisions, not only by arrival order.

T03 - Safeguarded processing flow

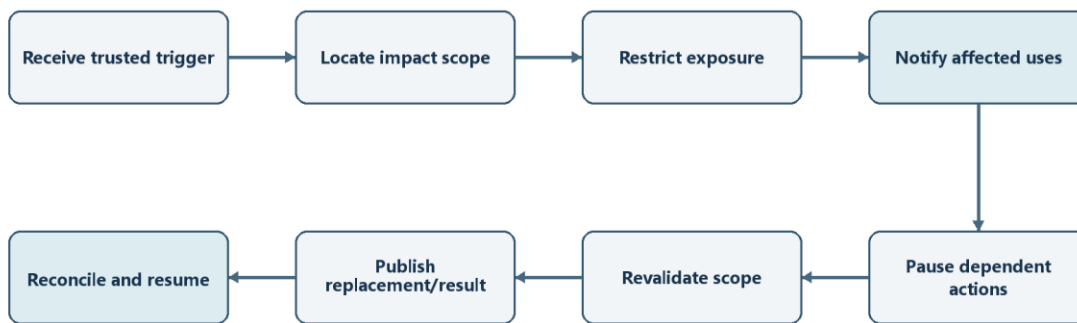


Figure T03-1. Logical interaction and safeguarded processing flow.

2.3 Disconnection, retention and use-time confirmation

Notification delivery is not assumed to be reliable or immediate. A consumer may query the current status of its provision/consumption reference after reconnection. Where a bounded-use profile is applied, the result includes an expiry or recheck condition; expiry limits the time for which old evidence may be used, but does not override an earlier invalidation.

For a critical action, a consumer may request confirmation for a specific revision, purpose and required freshness immediately before execution. The producer either confirms an applicable revision within the supported interval, provides a replacement, or returns unconfirmed. A consumer cannot infer confirmation from timeout. The

confirmation is bounded by its stated conditions; it cannot guarantee that the network will remain unchanged after it is issued.

If dependency records have expired, the producer conservatively widens the affected scope or reports that selective impact cannot be established. Retention of consumption information follows authorization and data-minimization policy. A deleted source does not require indefinite retention of its full content merely to support an impact reference.

2.4 Example and evaluation

A topology relation used for cell energy optimization is corrected. Consumers that only used an unrelated performance trend need not be interrupted if independence is established. A consumer preparing a switch-off action based on the corrected relation pauses and rechecks. If the producer cannot establish which results depended on that relation, it invalidates the broader instance instead of silently missing the consumer.

Tests include missed notifications, late restore messages, incomplete dependency evidence, changed consumer purpose and actions already executed. Evaluation compares global invalidation and selective invalidation by missed affected uses, unnecessary interruption, recovery delay and retained dependency volume. The proposal does not claim a measured reduction.

3. Impact, references and proposed agreement

Potential additions concern affected-scope, revision and recovery semantics where existing notification/result capabilities are insufficient. No complete lineage KI is reintroduced; no automatic reversal of all consumer actions is required. Sources: S5-263545, S5-263546 and S5-263547. PROV-O and incremental validation literature are relevant implementation background.

SA5 is invited to include this candidate solution and to align its impact semantics with knowledge states and contribution validation, keeping their respective responsibilities distinct.

T04 Contextual conflict detection and unresolved-result handling

Document status: complete technical draft. Target: TR 32.801-01, Knowledge Conflict Handling. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263546 retains conflict detection while explicitly leaving handling of unresolved conflicts and possible MnS effects for further study. Individual assertions can each be valid but jointly incompatible in the same context. Automatically preferring the latest contribution or the most frequent answer is not adequate for all management knowledge. This contribution provides a minimum safe detection and unresolved-result procedure without standardizing every resolution policy.

2. Proposed text: contextual detection and safe isolation

2.1 Conflict model

A conflict exists for this solution when a set of assertions, after entity and context alignment, cannot jointly satisfy an applicable constraint in an overlapping applicability domain. The domain may include managed entities, geographic or administrative scope, valid time and operating conditions. Different applicability domains may allow assertions to coexist.

The conflict record identifies the relevant assertions or authorized references, the overlapping context, violated constraint reference, evidence versions and detection status. It also identifies whether the conflict is in an unpublished candidate or affects already published knowledge. An unsupported comparison is reported as unresolved uncertainty, not as proof that the assertions are consistent.

2.2 Detection and isolation procedure

- Step 1. The producer receives a contribution or a trigger affecting existing knowledge, such as a source correction, changed rule or changed applicability context.
- Step 2. It normalizes entity references and applicable vocabulary according to supported mappings. Ambiguous identity or incompatible semantic profiles cause a request for clarification or an indeterminate result; assertions are not combined using guessed equivalence.
- Step 3. It identifies assertions with overlapping applicability and constraints potentially affected by the change. Direct content overlap is one condition, but shared constraints may require comparing otherwise different assertions.
- Step 4. It evaluates whether the assertions can jointly satisfy the constraints. It records the conflicting set and the context in which incompatibility arises. A conservative superset is allowed when a minimal conflicting set cannot be established, provided the result does not falsely claim minimality.
- Step 5. If the conflict concerns an unpublished candidate, the candidate or affected atomic contribution is not published. If previously exposed knowledge is affected, the producer applies suspected-invalid handling and identifies affected previous consumers using the validation procedure.
- Step 6. The producer returns an unresolved-conflict result including authorized scope, reason categories and references for follow-up. It may redact protected source details while still explaining that joint validity could not be established.

- Step 7. A contributor or authorized resolution function supplies corrected context, replacement assertions, additional evidence or an authorized policy decision. The old conflict record remains associated with the subsequent handling result for the supported retention period.
- Step 8. The producer validates the resulting candidate and current context before releasing the restriction or publishing a replacement revision. A proposed resolution is not valid merely because it came from a preferred contributor.

T04 - Safeguarded processing flow

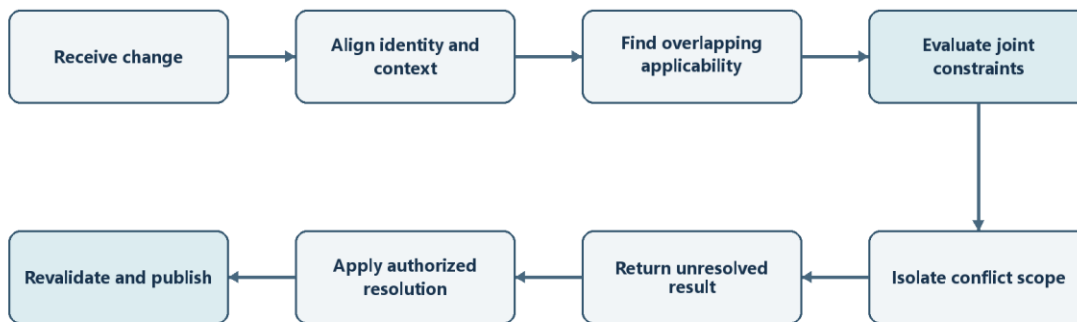


Figure T04-1. Logical interaction and safeguarded processing flow.

2.3 Permitted resolution options and boundaries

The following options may be evaluated without making one mandatory algorithm: retain both assertions under demonstrably different contexts; replace a superseded assertion with evidence of supersession; apply an authorized source-priority rule within its declared domain; obtain additional evidence; or request human resolution. The chosen policy and its applicability are recorded.

If no authorized policy can resolve the conflict, it remains unresolved. A consumer receives either unaffected knowledge with a clear scope boundary, or a result indicating that the requested valid knowledge cannot be provided. Independent validity of the unaffected scope must be established. If it cannot be established, isolation applies to the whole instance rather than to a convenient but unsafe subset.

Confidence and timestamps may be evidence for a specific policy. They are not universal replacements for semantic constraints. The producer does not expose one arbitrary member of a conflict as valid simply because it is newer or more popular.

2.4 Information requirements and failure cases

Required semantics are conflict correlation, affected revision/scope, applicability overlap, check status, resolution authorization and post-resolution validation result. These may be carried by existing result/notification capabilities where sufficient; this study does not decide a new conflict-management MnS.

If a source disappears during investigation, the conflict remains unresolved unless remaining evidence is sufficient. If a policy changes while a resolution is being validated, applicability is checked again before publication. A duplicate correction request resolves to the prior result when its content and correlation match; changed content starts a new candidate.

2.5 Worked example and evaluation

An assertion states that a cell can be disabled during a low-load interval. Another requires that the same cell remain active to maintain a service coverage condition. If the service requirement applies only after 06:00 and the energy-saving assertion ends at 05:00, both can coexist. If both apply at 04:00, the producer isolates the incompatible combination until context or evidence is corrected. A producer identity or arrival timestamp alone is not a sufficient resolution rule.

Evaluation covers context-disjoint assertions, combined constraints, identity ambiguity, rule changes after publication and incomplete evidence. Criteria are missed conflicts, over-isolation, explanation sufficiency and repeatability of policy application. Resolution effectiveness is evaluated separately from basic detection accuracy.

3. Impact, references and proposed agreement

Sources are the final S5-263546 conflict KI and its Editor's note, together with S5-263545/263547 for related states and architecture. The proposed text supplies a candidate answer for unresolved-result behavior without removing the need to study other resolution mechanisms.

SA5 is invited to include this solution and evaluate which information is already expressible by existing knowledge contribution, retrieval and notification capabilities before defining interface changes.

T05 Cause-aware knowledge lifecycle transitions

Document status: complete technical draft. Target: TR 32.801-01, Knowledge Lifecycle Management. Related issues: validation and knowledge provision. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263545 retains ACTIVE, SUSPENDED and DELETED while leaving transition conditions for further study. SUSPENDED can cover access unavailability as well as suspected invalidity. These causes have different implications: a temporary retrieval outage need not prove previously provided content invalid, whereas suspected invalidity requires protection of both new retrieval and relevant previous use.

This contribution describes transition conditions and their observable consequences without introducing CREATED or a large new state enumeration. It also separates lifecycle state from contribution-job progress and health assessment.

2. Proposed text: cause-aware lifecycle control

2.1 State and associated condition

ACTIVE denotes knowledge that is available for the supported provision operations and satisfies the applicable validity conditions. SUSPENDED denotes knowledge whose provision is restricted because an applicable condition is not satisfied or cannot be confirmed. DELETED denotes knowledge that is no longer accessible as the referenced instance under this lifecycle.

A state change is associated with a reason category, affected revision/scope, effective time, and a recovery condition or result reference where available. Suggested reason categories for study are access unavailable, suspected invalid, administrative restriction and unsupported recovery evidence. These are explanatory information, not mandatory new top-level states.

Lifecycle state does not replace authorization checks. ACTIVE knowledge may still be inaccessible to an unauthorized consumer. Similarly, a healthy-looking characteristic does not independently return suspended knowledge to ACTIVE.

2.2 Transition conditions

Transition	Condition and consequence
ACTIVE to SUSPENDED	An access, validity or authorized administrative condition is no longer satisfied. New provision is restricted for the affected scope.
SUSPENDED to ACTIVE	The condition causing suspension has been cleared and applicable validity and access checks succeed for the current revision.
ACTIVE or SUSPENDED to DELETED	An authorized deletion is committed; ordinary retrieval is no longer possible. Retained audit references do not expose deleted content.
DELETED to renewed provision	No implicit reversal is assumed. Renewal creates a new instance or follows an explicitly supported revision/re-creation rule that preserves unambiguous identity.

Where the knowledge model does not support independent lifecycle control of subscopes, these rules apply to the whole instance. A producer must not label an instance ACTIVE while silently treating an inseparable portion as invalid.

2.3 Procedure

- Step 1. The producer detects or receives an authorized event that can affect access or validity. It determines the affected revision and distinguishes operational unavailability from evidence of suspected invalidity.
- Step 2. It records the event and computes the scope that can be controlled independently. If that scope cannot be determined reliably, it uses a broader lifecycle unit.
- Step 3. It applies the state transition and associated reason before providing further affected content. A pending retrieval either completes using a still-applicable validated revision or returns a failure/status change; it does not silently cross the state transition.
- Step 4. For access-only unavailability, it reports that new retrieval is unavailable. Previously delivered knowledge is not automatically declared invalid, but remains subject to its validity, expiry and authorization conditions.
- Step 5. For suspected invalidity, it invokes the validation impact procedure, prevents affected exposure and informs identifiable previous consumers. Consumers check related pending actions and obtain confirmation or replacement knowledge.
- Step 6. On a recovery event, it checks the cause-specific condition. Restored connectivity addresses an access failure, but does not by itself clear a suspected-invalid condition. The latter requires applicable revalidation evidence.
- Step 7. It atomically records the resulting state/revision association and returns or notifies the recovery result through supported capabilities. Late notifications for older revisions cannot override a later deletion or unresolved invalidation.
- Step 8. On deletion, it terminates ordinary access and maintains only the authorized minimal operation/audit references required for result reconciliation and retention obligations.

T05 - Safeguarded processing flow

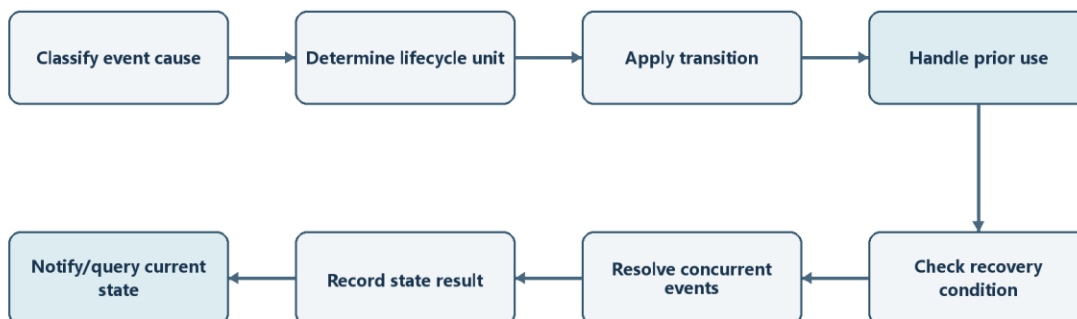


Figure T05-1. Logical interaction and safeguarded processing flow.

2.4 Bounded use and concurrent events

For use cases requiring bounded exposure to stale decisions, knowledge provision may include an expiry or a use-time recheck condition bound to the revision and purpose. A consumer stops using the knowledge when the condition fails, even if it has not received an invalidation notification. An earlier invalidation takes precedence over an unexpired permission.

When access restoration and a source invalidation occur concurrently, the producer evaluates both conditions. Restoration clears only the access reason; knowledge remains suspended if validity remains unresolved. When deletion races with recovery, the committed deletion result is not undone by a late recovery message. A consumer that cannot determine the order queries current state rather than choosing the last-arriving message.

2.5 Example and evaluation

A producer temporarily loses access to its retrieval backend while a consumer holds a recently validated forecast. The consumer can use the forecast within its stated conditions if no contrary validity evidence exists. In a separate event, the underlying measurement is corrected; the producer marks affected knowledge suspected invalid, and the same consumer must reevaluate use even if the original forecast expiry has not passed.

Evaluation should cover overlapping suspension reasons, access restoration without validity restoration, deletion during revalidation, and notification reordering. Relevant outcomes are improper reactivation, unintended continued provision, unnecessary consumer interruption and recoverability. The solution intentionally avoids prescribing internal database states or transition algorithms.

3. Impact, references and proposed agreement

Sources: S5-263545 knowledge states and S5-263546 validation requirements. Potential additions are reason, effective scope/revision and recovery semantics where absent from existing capabilities. This proposal does not restore the deleted independent Health-change notification requirement; it uses lifecycle/validation behavior appropriate to the cause.

SA5 is invited to include these transition principles as a candidate solution and evaluate the supported granularity of state and validation information together with the knowledge model.

T06 Revision-consistent knowledge query via context

Document status: complete technical draft. Target: TR 32.801-01, Knowledge Provision / Knowledge Query via context. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263545 retains query via context and leaves aspects of the baseline for further study. Detailed SPARQL, namespace and serialization-discovery mechanisms were removed. The remaining question is not which query language must be used, but how a consumer can interpret context filtering, result completeness and changes during retrieval.

A large result may span pages while the knowledge or its semantic mapping changes. A subscription started after a snapshot may miss changes at the handover. This contribution defines consistent result semantics and an optional incremental extension, independent of wire syntax and internal query technology.

2. Proposed text: context query with identifiable revision

2.1 Query and result semantics

A query identifies the requested scope, applicable context, required freshness or other assessable acceptance conditions, and completeness preference. Context may refer to network entities, a geographical or management domain, a time interval, and relevant generation information. The producer checks that the requested conditions are supported and authorized before evaluating them.

The result identifies the knowledge revision or consistent revision set, semantic profile version, evaluation time, provided scope, validation applicability and completeness. “No matching knowledge”, “not authorized”, “not evaluated”, “partial result” and “temporarily unavailable” have different meanings and are not collapsed into an empty successful response.

Information	Meaning
Query correlation and context reference	Identifies the query conditions used for result construction.
Result revision and semantic profile	Identifies the content and interpretation pinned for the result.
Continuation reference	Continues the same result view; it is not permission to switch silently to a newer revision.
Completeness and coverage	States the evaluated/provided scope and any unassessed or excluded part.
Retention limit	States until when the pinned view or continuation remains supported.

2.2 Snapshot query procedure

- Step 1. The consumer sends a context query and states whether partial results are acceptable. The producer checks authorization, supported context conditions and requested result limits.
- Step 2. The producer selects an available published revision or a consistent version set. It binds the query to the semantic mapping/profile required for interpretation and checks validity for exposure.

- Step 3. It evaluates the context and acceptance conditions. Unknown characteristic values remain unknown; they are not treated as satisfaction of a threshold. If the request disallows unassessed conditions, the corresponding result is not returned as compliant.
- Step 4. It returns the first result page with revision, profile, coverage and a continuation reference when needed. A continuation is scoped to the authorized consumer and result conditions.
- Step 5. The consumer retrieves further pages using the continuation. The producer uses the same pinned view and rechecks applicable authorization and invalidation conditions. It does not replace withdrawn knowledge with a new revision inside the old pagination sequence.
- Step 6. If the pinned view becomes unavailable, invalid or outside retention, the producer returns an explicit restart or unavailable result. The consumer discards an incomplete view or retains it only as an explicitly incomplete, unusable-for-the-request result.
- Step 7. On completion, the consumer can identify the complete revision/context it received. It may start a new query for a newer revision without confusing the two result sets.

T06 - Safeguarded processing flow

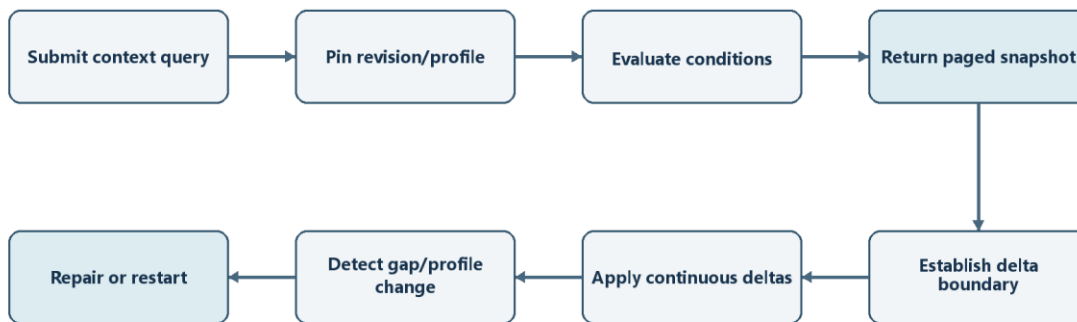


Figure T06-1. Logical interaction and safeguarded processing flow.

2.3 Optional snapshot-to-incremental procedure

The producer may offer an incremental continuation of a completed baseline. The starting reference identifies the baseline revision and context; the producer establishes the corresponding change boundary before reporting the incremental continuation as active. Where no reliable change boundary is available, it does not claim gap-free handover and instead requests a fresh baseline.

Each increment identifies its predecessor and resulting revision, plus additions, revisions and withdrawals within the result context. A relation change outside the original entity set can affect context membership or derived results and is included when required by the query semantics. The consumer checks predecessor continuity and applies only increments belonging to the same interpretation context.

On a missing increment, the consumer requests recovery. The producer provides a provably complete repair of the affected result scope, or a full snapshot if it cannot establish a safe partial boundary. A semantic-profile change that alters interpretation is handled as a new baseline or an explicitly validated transformation, not as an ordinary content delta.

2.4 Security, failure and example

Authorization is rechecked for continuing provision. Revocation stops further exposure and produces withdrawal or status information as permitted; it cannot guarantee erasure of content already copied by a consumer. A continuation reference is not an unrestricted bearer credential for unrelated knowledge.

For example, a consumer requests valid energy-saving recommendations for cells in a region. During pagination, one cell is reassigned to another region and a shared coverage relation changes. A pinned snapshot preserves one coherent initial result. An incremental continuation later removes or updates affected recommendations rather than merely forwarding field updates for cells currently listed in the cache.

Evaluation covers pagination during updates, expiry of pinned revisions, missing changes, context membership changes, profile upgrades and authorization revocation. Criteria are missing/duplicate result elements, unnoticed mixed revisions, incorrect retention of withdrawn recommendations and recovery traffic. Snapshot and delta algorithms remain implementation specific.

3. Impact, references and proposed agreement

The proposal does not mandate SPARQL, a graph query protocol, a new notification service or a specific storage system. Existing query/result/notification capabilities should carry the described semantics when possible. Sources: S5-263545 and S5-263547; HTTP and versioned API concepts provide background examples only.

SA5 is invited to include revision-consistent query semantics as a candidate solution and study incremental continuation as an optional extension with explicit continuity and restart behavior.

T07 Knowledge health evidence and purpose-specific assessment

Document status: complete technical draft. Target: TR 32.801-01, Knowledge Health Information Management.
Related issue: Knowledge Validation. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263546 retains creation/maintenance, request and provision of knowledge health information while leaving its relationship with validation for assessment. The meeting discussion questioned whether health could be measured and whether it overlaps lifecycle management. A single opaque health score would not answer these questions. The same valid knowledge may be suitable for long-term trend analysis and unsuitable for an immediate control action.

This contribution separates evidence about knowledge characteristics, assessment against a consumer's purpose and the producer's decision on validity for exposure. It provides two compatible responsibility allocations rather than a mandatory universal scoring formula.

2. Proposed text: evidence-based health information

2.1 Characteristic evidence

Health information describes characteristics of an identifiable knowledge revision and scope. Depending on the characteristic, the value may be numeric, categorical or a supported description. Evidence includes the assessment time, applicable coverage, method/profile reference and the status of availability. Immutable origin information and characteristics that change with time are distinguished when interpreting freshness.

Characteristic example	Evidence interpretation
Freshness	Age relative to an identified source/assessment time and a stated reference clock or uncertainty.
Completeness	Evaluated content relative to an identified expected set, not an unexplained percentage.
Coverage	Managed entities, context or time interval to which the characteristic applies.
Consistency	Result and scope of the specified consistency checks, not a claim that every possible constraint was evaluated.
Origin/generation information	Supported references explaining source and generation applicability, subject to authorization.

Unknown, not applicable and not evaluated are distinct. If an expected set is not available, the producer does not fabricate a completeness ratio. If a characteristic method changes, its version accompanies the result so that consumers do not compare incompatible measurements as if they were identical.

2.2 Option A: consumer-side purpose assessment

- Step 1. The producer creates or updates characteristic evidence for a knowledge revision as relevant inputs or assessment times change.
- Step 2. A consumer requests the knowledge and associated health information, or health information for an authorized existing revision reference.

- Step 3. The producer checks authorization and validity conditions for knowledge exposure. A failed validity gate cannot be bypassed by supplying favourable health evidence. Restricted diagnostic health information may be provided separately if authorized and clearly labelled.
- Step 4. The producer returns supported characteristics, evaluated coverage, evidence time and unknown/not-evaluated indicators. It does not infer the consumer's unspecified operating purpose.
- Step 5. The consumer applies its purpose conditions to that evidence. It records whether the provided knowledge is acceptable, unacceptable or unassessed for that purpose and retains the revision/profile association needed to reproduce the decision.
- Step 6. When the conditions or evidence age no longer satisfy the purpose, the consumer requests updated information or a new knowledge result. This does not by itself redefine the producer's global validity state.

2.3 Option B: producer-side assessment against supplied conditions

The consumer may instead supply a purpose profile expressed as assessable conditions: target scope, maximum source age, expected coverage and other supported acceptance predicates. The producer confirms that the conditions are understood, evaluates them against authorized evidence, and returns satisfied, not satisfied or unknown for each condition, together with the overall assessment rule used.

A conservative overall rule is that all mandatory conditions must be satisfied, any failed mandatory condition makes the result unsuitable, and otherwise a mandatory unknown produces an unassessed result. Other supported rules must be explicit. Optional conditions are reported separately and cannot silently cancel a failed mandatory condition. The result is bound to the purpose profile, knowledge revision and assessment time.

T07 - Safeguarded processing flow

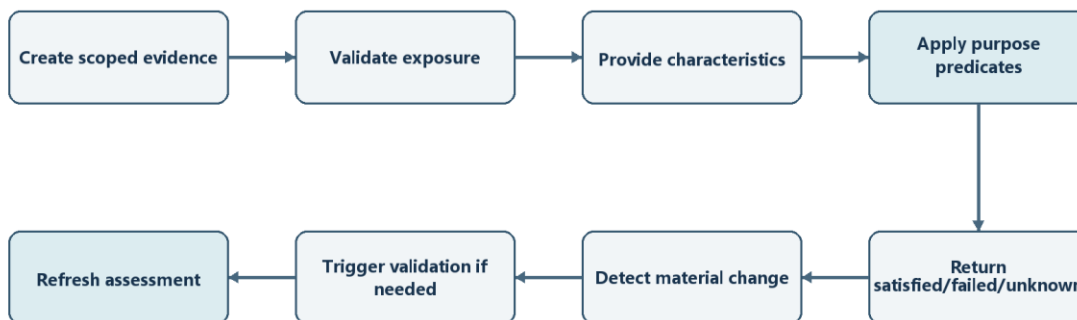


Figure T07-1. Logical interaction and safeguarded processing flow.

2.4 Relationship with validation and lifecycle

Validation determines whether knowledge is valid for the applicable exposure context. Health information provides evidence that may be used by such validation or by a consumer's suitability assessment. Lifecycle state controls availability and applicable state transitions. The same evidence can be reused without requiring the three concepts to have identical outcomes.

If an assessment reveals a possible validity defect, it can trigger the existing validation investigation and suspected-invalid handling. A purpose-specific failure such as “too old for immediate control” need not invalidate

the knowledge for every purpose. No new independent health-change notification is assumed; changes are retrieved through supported capabilities or trigger validation behavior when relevant.

2.5 Worked example, limitations and evaluation

Knowledge based on measurements from twenty minutes ago may satisfy a trend-analysis profile allowing one hour of age, but fail an immediate-control profile allowing one minute. These numbers are illustrative configuration values, not proposed standards or measured thresholds. If the producer cannot establish the source time, both profiles receive unknown freshness rather than an age of zero.

Evidence can be expensive to compute. A producer may support a subset of characteristics or cached assessments, provided the assessed scope and time are explicit. The solution does not require disclosure of proprietary scoring algorithms; however, a consumer must receive enough semantics to understand the reported characteristic and its limitations.

Evaluation covers unknown values, conflicting purpose conditions, changed expected coverage, aged assessments and simultaneous validation failure. Criteria are reproducibility, inappropriate cross-purpose reuse, misleading numeric values and exposure of invalid knowledge. No universal health ranking is proposed.

3. Impact, references and proposed agreement

Source: S5-263546, Knowledge Health Information Management and Knowledge Validation. Potential information-model work concerns evidence scope/time, characteristic status and purpose-assessment binding. No new top-level health state machine or mandatory score is introduced.

SA5 is invited to include both responsibility options for study, agree on the conceptual separation above, and evaluate minimal interoperable evidence before selecting any concrete representation.

T08 Mapping provisioning requests to declarative deployment outcomes

Document status: complete technical draft. Target: TR 32.801-01, Integration of declarative-based LCM with 3GPP SBMA. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263625 retains study of desired state determined through standard management services, mapping between NF operations and NF Deployment actions, progress/result feedback and desired-state modelling. The earlier DeclarativeLcmMnS solution was removed. S5-263621 keeps the approach technology neutral and identifies open distinctions between NF and NF Deployment desired state.

A cloud controller reporting success for a deployment revision does not necessarily complete the corresponding 3GPP request. The NF may still need configuration, the feedback may concern an older desired revision, or the service acceptance condition may not hold. This contribution defines correlation and completion semantics without prescribing Git or a new management service.

2. Proposed text: request, desired revision and outcome correlation

2.1 Roles and correlated information

A 3GPP management service consumer submits a provisioning request to the responsible producer. That producer determines NF-related changes and the required NF Deployment target. An external orchestration/management system executes deployment changes through an applicable reference point. The 3GPP producer remains responsible for reporting the result of the original management request according to its service semantics.

Correlation element	Purpose
Management request reference	Identifies the original request and its authorized scope.
Desired-state revision	Identifies the target that a deployment result is expected to realize.
External execution reference	Distinguishes execution attempts, retries and externally reported operations.
Observed revision and state	States which target revision the external report actually concerns.
NF configuration and acceptance evidence	Determines whether deployment success also satisfies the NF/service conditions relevant to the request.

These are logical information requirements. They may be derived from existing operation references or versions, and are not a decision to add all fields to every interface.

2.2 Procedure

- Step 1. The consumer issues an authorized provisioning request. The producer records the request reference, target NF scope and relevant preconditions.
- Step 2. The producer determines the NF configuration change and associated NF Deployment target. It creates or identifies a desired revision *d* and records how the management request depends on *d*.

Changes that do not require deployment actions remain within the applicable NF management procedure.

- Step 3. The producer verifies that the external system can express and apply the required target and report sufficient result information. Unsupported constraints are rejected or reported for resolution, not silently discarded.
- Step 4. The producer requests realization of d through the applicable external capability and records the execution reference. Receipt by the external system is reported only as receipt/progress for the original request.
- Step 5. External progress is correlated to d and the execution attempt. A result for an older desired revision may update its historical operation record but cannot complete a request depending on a newer revision.
- Step 6. Once deployment readiness for d is established, the producer performs or confirms applicable NF configuration and other acceptance checks. It distinguishes deployment readiness from NF readiness and from any service-level condition required for the request.
- Step 7. The producer reports success only when the current request's applicable conditions are satisfied. It reports in-progress, failure or an unresolved outcome when the corresponding evidence does not support success.
- Step 8. The consumer can retrieve the correlated result for audit and recovery. Repeated or reordered feedback does not create repeated logical completions.

T08 - Safeguarded processing flow

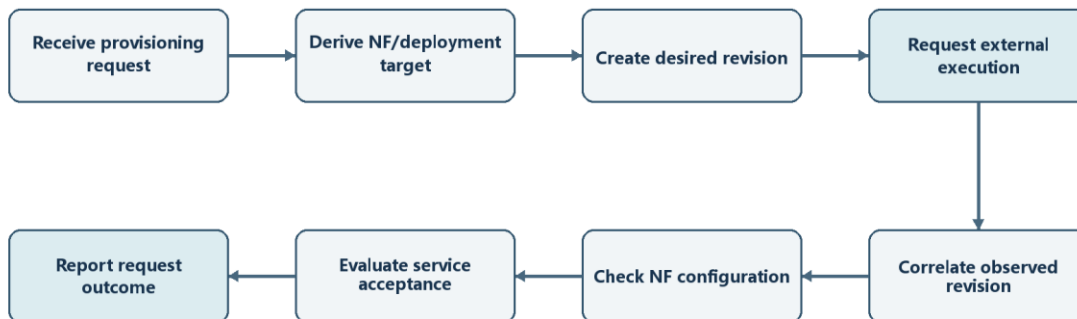


Figure T08-1. Logical interaction and safeguarded processing flow.

2.3 Desired-state replacement and external change

If a new authorized request replaces d with $d+1$, the producer records which earlier requirements remain applicable and whether an earlier operation is cancelled, superseded or independently completed. It does not infer these semantics solely from the external controller's latest state. A late success for d is processed against the earlier operation and any explicitly retained dependency.

An external system may change a deployment autonomously within its authorized scope. Such actual-state feedback can trigger reconciliation or a new management decision, but does not automatically create a new authorized 3GPP desired state. Unexplained drift is reported as a difference between actual state and the applicable target.

If the external system cannot report a revision directly, the producer may correlate a stable operation reference with a separately queried actual configuration. If neither method can establish that the target was realized, the result remains unconfirmed rather than being assigned success on timing alone.

2.4 Failure and example

A timeout does not prove external failure. The producer queries the external operation and actual state before deciding whether a retry is safe. Cancellation of the consumer request does not prove that deployment activity stopped. The associated operation remains tracked until its consequences are known or explicitly reported as unresolved.

For example, request R1 requires revision d1 and request R2 later requires d2. A success callback for d1 arrives after d2 is submitted. R2 stays in progress. If deployment d2 becomes ready but NF configuration fails, R2 is not successful even though the cloud controller has converged. Partial failure is handled using a separately defined recovery procedure.

Evaluation covers late success, duplicate feedback, target replacement, autonomous drift, unsupported revision reporting and deployment/configuration mismatch. Criteria include incorrect success, lost correlation and recovery after an uncertain external result. Kubernetes observedGeneration is relevant background but does not itself supply all 3GPP request-completion semantics.

3. Impact, references and proposed agreement

Sources: S5-263621, S5-263625 and S5-263620, with TS 28.531/28.533 as existing framework/provisioning references. No new DeclarativeLcmMnS, Git repository, YAML/Helm encoding or cloud-controller implementation is required.

SA5 is invited to include the solution and assess the minimal correlation and acceptance information that existing management services and external reference points can support.

T09 Partial failure and safe compensation in declarative LCM

Document status: complete technical draft. Target: TR 32.801-01, declarative-based LCM integration. Related issue: sub-network LCM. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

The feedback and audit aspects retained in S5-263625 require more than a final success/failure label. Deployment, NF configuration and service acceptance may complete at different times. Undoing every earlier change after a later failure can damage another operation that has already reused the resource. Conversely, reporting cancellation while an external action continues leaves the actual network state unclear.

This contribution defines a staged recovery procedure using correlated operations and actual-state reconciliation. It does not require a distributed atomic transaction across the 3GPP producer and external cloud systems, and it does not assume every deployment action has an inverse operation.

2. Proposed text: staged acceptance and dependency-aware recovery

2.1 Processing record and conditions

The responsible producer maintains a logical processing record for a management request. It identifies the desired revision, external operation references, affected objects, completed acceptance stages and supported recovery actions. Stages may include prerequisite checking, deployment readiness, NF configuration and service acceptance. Implementations need not use identical internal state names.

For each change, the record distinguishes the object state observed before the change, the modification attributable to this request, and any later dependencies known to the management system. The record does not claim exclusive ownership of externally managed resources merely because this request created them.

Recovery condition	Permitted handling
Target can still be reached safely	Continue reconciliation or retry an identified failed stage.
Change is reversible and has no conflicting later dependency	Request compensating change through the authorized capability.
Later requests/services depend on the changed resource	Preserve it or use an alternative recovery that maintains those dependencies.
External result or reversibility is unknown	Query, pause or request authorized manual handling; do not assume rollback succeeded.

2.2 Normal and failure procedure

- Step 1. Before an external action, the producer records the logical request, desired revision, intended effect and relevant preconditions. If the external capability supports an idempotency key or operation lookup, the producer binds it to that record.
- Step 2. The producer executes applicable stages, collecting evidence for each. Completion of one stage does not imply completion of later stages. Actual external references replace provisional planning references as soon as they are known.

- Step 3. A stage fails, times out, is cancelled or becomes superseded. The producer stops issuing incompatible new actions, while continuing to observe already issued external actions.
- Step 4. The producer reconciles actual state and outstanding operations. It identifies which effects definitely occurred, which are still executing and which remain unknown. A timeout is not classified as “no effect”.
- Step 5. For each known effect, the producer checks its relationship to the current desired target and later authorized operations. The check covers shared NF Deployments, configuration references, service associations and other relevant dependencies available to the responsible function.
- Step 6. It chooses forward recovery, safe compensation, preservation with partial-failure reporting, or escalation. A compensating request includes appropriate state/revision preconditions where supported so that a concurrent later change is not overwritten.
- Step 7. The external system executes the authorized recovery action and reports its result. The producer verifies resulting state and dependent acceptance conditions; an acknowledgement of the compensating request alone does not prove successful compensation.
- Step 8. The producer reports the original outcome and recovery outcome separately, including remaining effects and unresolved conditions. It maintains a stable audit association rather than rewriting a failed request into an unqualified success.

T09 - Safeguarded processing flow

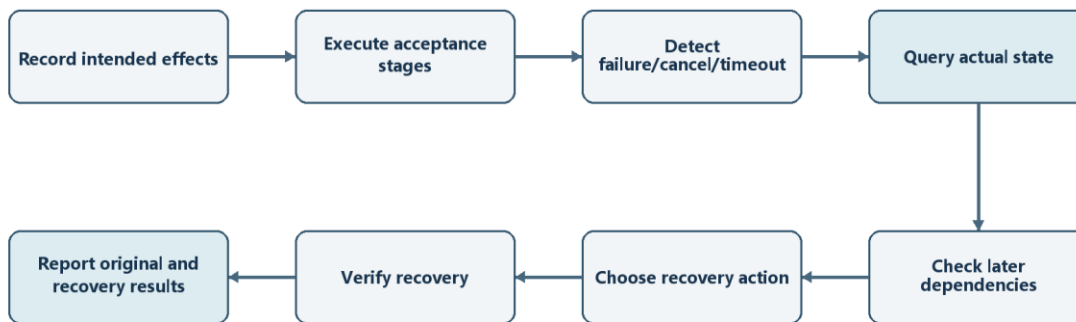


Figure T09-1. Logical interaction and safeguarded processing flow.

2.3 Concurrency and unsupported capabilities

If a dependency changes between planning and executing compensation, the producer reevaluates the action or uses conditional execution to detect the change. If an external capability does not offer conditional execution, the producer applies a supported serialization/coordination arrangement or reports that safe automatic compensation cannot be established.

Cancellation is an intent to stop the requested activity; the result states whether external execution was prevented, stopped after partial effects, completed before cancellation, or remains unknown. A late successful external response updates the corresponding operation and may trigger further reconciliation. It does not resurrect a cancelled desired target without authorization.

Operations with irreversible effects, such as deletion without recoverable state, are not described as automatically reversible. Such actions require an applicable policy and prerequisites before execution. A generic compensation framework cannot restore data or service state that was never retained.

2.4 Worked example

R1 deploys a new NF instance and begins configuring it. R2 subsequently establishes a service association that legitimately depends on the deployment. R1 then fails a remaining configuration check. A naive rollback deletes the deployment and breaks R2. Under this solution, the dependency check prevents deletion; the producer repairs the failed configuration if possible, or preserves the deployment while reporting the incomplete R1 result and affected scope.

In a second branch, no later operation depends on the deployment and its deletion is authorized and reversible within the applicable requirements. The producer requests removal, verifies actual removal and reports both R1 failure and successful cleanup. The two branches use the same error trigger but produce different safe actions.

2.5 Evaluation and interface implications

Tests cover lost external responses, stage failures, overlapping requests, cancellation during deployment, concurrent dependency creation and failure of compensation itself. Metrics are unintended removal of dependent resources, duplicate external effects, unresolved-state duration and incorrect final reporting. No exactly-once external side-effect guarantee is asserted.

The potential standardization requirement is observable result and recovery semantics. The internal workflow engine, dependency graph and recovery optimizer remain implementation choices. Existing operation/result and provisioning capabilities are reused when they can express the necessary preconditions and outcomes.

3. References and proposed agreement

Sources: S5-263625, S5-263622 and the existing TS 28.531/28.533 framework. Request/revision correlation is compatible with T08 but the present text defines its own processing prerequisites and can be evaluated independently.

SA5 is invited to include this candidate solution and study partial-effect, cancellation and recovery-result semantics as part of declarative LCM integration.

T10 Temporal association of NF, deployment and runtime observability

Document status: complete technical draft. Target: TR 32.801-01, Network Functions Observability. Related issue: NRM improvements for NF Deployments. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263623 retains cross-NF/domain correlation and industry-standard integration, while the earlier agent/orchestrator-specific solution was not retained. S5-263622 leaves NF Deployment modelling open. During rolling replacement and migration, current runtime-to-NF mapping can be wrong for historical events. Reused names and clock differences make a single present-time identifier insufficient.

This contribution addresses association semantics without requiring a new observability agent, a particular telemetry transport or a predetermined NF Deployment IOC. It applies to metrics, logs and traces and allows physical network functions without a cloud runtime entity.

2. Proposed text: temporally qualified association

2.1 Association information

A logical NF identity, a deployment identity and a runtime entity identity are distinct. An association relates applicable identities for a validity interval and identifies the authority/source of that association, its revision and any known uncertainty. Multiple runtime entities may be associated with one NF during replacement; the model does not assume fixed cardinality.

An observation provides event time, collection/arrival time where available, source identity and supported association references. A time uncertainty bound or an explicit unknown is retained when clocks cannot be assumed aligned. Arrival time is not silently substituted for event time in causal analysis.

Information	Purpose
Stable entity references	Distinguish logical function, deployment and runtime instance; names alone may be insufficient.
Association valid interval and revision	Describe which relation applied at the event time.
Observation event time and uncertainty	Support temporal matching and identify ambiguous boundary events.
Association confidence/status	Distinguish established, ambiguous and unavailable association evidence.
Analysis dependency reference	Identifies which conclusions depend on an association that may later be corrected.

2.2 Procedure

- Step 1.
- The responsible management capability obtains associations from authorized NF/deployment inventory and external runtime information. It records historical validity rather than overwriting every old relation with the current one.

- Step 2. Collection capabilities provide observations using existing supported mechanisms. The observability analysis receives available identity and timing information; missing fields are reported as limitations, not invented.
- Step 3. The analysis forms the event-time interval implied by the timestamp and uncertainty. It retrieves associations whose valid interval and identity evidence are compatible with that event.
- Step 4. If exactly one association can be established for the required analysis granularity, the event is related to the corresponding NF/deployment context. If multiple relations remain possible, a candidate set is retained. If none can be established, the event remains unassociated for that purpose.
- Step 5. The analysis correlates observations across NF and infrastructure dependencies using established or explicitly uncertain relationships. An uncertain relationship cannot support an unqualified unique-root-cause claim.
- Step 6. A late inventory update or observation may correct an association interval. The analysis identifies dependent events and conclusions, recomputes the affected set and issues a correction or withdrawal of earlier conclusions where needed.
- Step 7. Consumers reconcile corrections by analysis-result reference and version. An old conclusion is not kept active merely because it arrived earlier or was already displayed.

T10 - Safeguarded processing flow

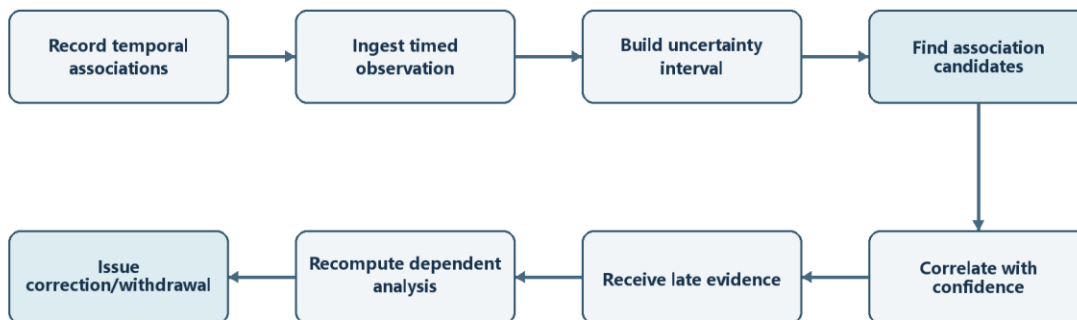


Figure T10-1. Logical interaction and safeguarded processing flow.

2.3 Deployment alternatives and retention

Association information may be materialized within an enhanced management model, or obtained through stable external references and an on-demand view. Either approach needs temporal and revision semantics sufficient for the intended use. This solution does not preselect a new IOC, retire an existing attribute or mandate a one-to-one relationship.

Retention is based on the supported observation analysis window and applicable policy. If an event predates retained association evidence, the analysis reports that historical association cannot be established. Guessing from the current mapping is not a valid recovery. A PNF can be represented using its logical/physical identity and applicable relation without a fabricated cloud deployment.

2.4 Worked example and failures

At a replacement boundary, runtime instance p-old stops serving an NF and p-new starts. A delayed log from p-old arrives after the inventory shows p-new. The analysis uses the event time and retained association interval to place the log in the old execution context. If timestamp uncertainty spans the handover and evidence cannot distinguish the instances, it retains both candidates and avoids a definite attribution.

If a deployment identifier is reused, an instance generation or equivalent stable external reference distinguishes the histories. If no such distinction is available, the analysis marks the association ambiguous. Loss of an external inventory service can reduce the available correlation scope, but must not silently improve confidence.

2.5 Evaluation and capability reuse

Evaluation scenarios include rolling upgrade, migration, simultaneous replicas, identifier reuse, delayed logs, clock uncertainty and mixed PNF/CNF service chains. Criteria are misassociation, false causal certainty, correction completeness and retention/index cost. Data collection and identity transport should reuse existing management and industry capabilities where possible.

The integration study can map OTel service/resource attributes to the required identity context. This does not mean OTel defines all 3GPP NF/deployment relationships or that the proposed association requires OTLP. Agent placement and internal indexing are outside the solution's required architecture.

3. References and proposed agreement

Sources: S5-263623 and S5-263622; OpenTelemetry service semantic conventions are external technical background. The proposal addresses correlation and modelling information needs together, to avoid independent incompatible identity schemes.

SA5 is invited to include this candidate solution and identify the minimal association information missing from existing NRM and observation mechanisms before selecting concrete model changes.

T11 Resource-bounded observability control with applied-coverage feedback

Document status: complete technical draft. Target: TR 32.801-01, Network Functions Observability, especially control/management and cross-domain analysis. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263623 retains observability control and industry integration. Meeting comments warned about overlap with DMFW and existing collection. A new general-purpose collector is therefore not the starting point. The gap studied here is a control request that asks for additional evidence but receives less than requested because of resource limits, unsupported probes or changing deployment conditions.

Analysis must know the actual coverage it obtained. Otherwise, “no observed fault evidence” can be mistaken for “fault absent”, even though the requested observation was never collected. This contribution combines bounded temporary control with explicit applied-coverage feedback.

2. Proposed text: evidence-oriented temporary observation control

2.1 Roles and information

An analysis capability identifies the evidence needed to distinguish plausible explanations of a management problem. A control capability coordinates authorized changes to existing collection functions. These are logical roles that may be provided by existing management functions; no mandatory new agent or orchestrator is introduced.

Information	Meaning
Observation objective and scope	Identifies the entities, time window and supported evidence types required for analysis.
Requested configuration	Describes sampling, granularity or selected measurements/traces using supported controls.
Resource and duration bounds	Limits additional CPU, bandwidth, storage or other assessable resource use and defines expiry.
Applied configuration and coverage	Reports what was actually enabled, when, where and with what known omissions.
Termination/reversion result	States whether temporary changes were removed or what difference remains.

The analysis may use an internal fault-hypothesis model, but that model need not be exposed or standardized in full. Requested objectives and actual coverage must nevertheless have interoperable meanings for the chosen control capability.

2.2 Procedure

- Step 1. The analysis identifies an unresolved problem and the evidence currently missing to distinguish its plausible causes. It identifies the relevant NF/deployment/runtime scope using available association information.

- Step 2. The control capability obtains supported observation controls and applicable resource limits through existing capability information or configuration. Unsupported evidence types are excluded or explicitly reported.
- Step 3. A bounded request is formed with scope, configuration, start/duration conditions, resource limits and a correlation reference. The request is authorized against the affected domains and protected data requirements.
- Step 4. The execution capability admits the request fully, partially or not at all. It returns the applied configuration, effective interval and scope. If it adapts the request to protect resources, the adaptation is observable rather than hidden.
- Step 5. Observations are collected using supported mechanisms. Collection metadata identifies coverage gaps, relevant sampling behavior and changes during the interval when known.
- Step 6. The analysis evaluates the observations together with actual coverage. A hypothesis requiring an uncollected trace is not eliminated simply because that trace is absent from the result.
- Step 7. If uncertainty remains, the analysis may request a revised bounded configuration, within the remaining budget and authorization. It does not accumulate unbounded temporary probes through repeated requests.
- Step 8. At completion, expiry or resource pressure, the execution capability removes or reduces temporary configuration and reports the final applied state. The analysis records whether sufficient evidence was obtained or the objective remained unresolved.

T11 - Safeguarded processing flow

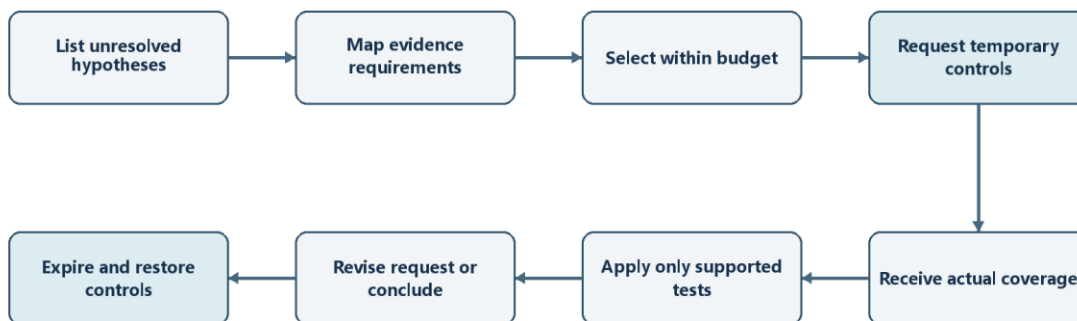


Figure T11-1. Logical interaction and safeguarded processing flow.

2.3 Multiple requests and resource pressure

Concurrent requests may share collection work when their scope and required semantics are compatible. The execution capability reports the actual effective configuration to each requester. It does not promise an isolated budget if resources are shared without accounting.

When available resources decrease, a predefined priority or admission policy may reduce coverage. The requester receives a change/result indication or can query the current configuration. If feedback is unavailable, the analysis treats coverage as unknown for conclusions requiring guaranteed observation. Expiry is enforced locally by the execution capability so that disconnection of the requester does not leave indefinite high-cost collection enabled.

Existing head/tail sampling or measurement controls may implement parts of this behavior. Their specific algorithms remain outside the proposed standard semantics. Security policy may prohibit payload or trace collection even when resource budget is available.

2.4 Example and evaluation

An NF exhibits latency. Possible causes are an application queue and a transport problem. Available metrics do not distinguish them. A request asks for a short interval of queue evidence and scoped traces under a bandwidth bound. The execution capability supports queue metrics but applies only partial trace coverage. The analysis can strengthen or reject the queue explanation as evidence permits, while retaining uncertainty about transport rather than declaring it healthy because trace evidence is missing.

Evaluation compares fixed sampling, unconditional escalation and bounded adaptive observation under identical resource budgets. Criteria are unresolved/incorrect diagnoses, time to distinguish causes, resource cost, accurate reporting of applied coverage and cleanup after requester failure. The contribution proposes no measured performance gain or mandatory optimization algorithm.

3. Impact, references and proposed agreement

Sources: S5-263623 and related SA5#168 comments on reuse of existing collection. OTel sampling concepts are background. Before defining an interface extension, SA5 should map requested/applied coverage, duration and budget semantics to existing DMFW, measurement and trace-control capabilities.

SA5 is invited to include this candidate solution and focus the gap analysis on applied-coverage feedback and bounded control, rather than assuming a new observability architecture.

T12 Sub-network LCM coordination using external NF deployment management

Document status: complete technical draft. Target: TR 32.801-01, Improvements to sub-network LCM based on NF Deployments. Related issues: Cloud framework and declarative integration. Official metadata is to be completed by the submitting organization.

1. Reason for contribution

S5-263622 retains high-level sub-network creation/configuration work based on NF Deployments, while detailed procedures and specific deployment parameter modelling were reduced. S5-263620 and S5-263624 point back to existing provisioning and architectural framework responsibilities. A complete candidate procedure should explain how external deployment execution contributes to a sub-network result without making the 3GPP system an infrastructure scheduler.

This contribution develops a process-oriented coordination path and compares a declarative delegation option. Both preserve the separation between deployment realization, NF configuration and sub-network acceptance.

2. Proposed text: sub-network lifecycle coordination

2.1 Roles, preconditions and information

A consumer requests creation or modification of a sub-network through the applicable 3GPP management capability. The responsible producer determines the required NF composition and configuration. External orchestration/management capabilities realize applicable NF Deployment requirements. Existing functions can implement these roles; this solution does not introduce a mandatory new sub-network cloud manager.

The producer needs the requested functional composition, constraints, existing NF/deployment associations, external capability references and acceptance criteria applicable to the request. Capability discovery or configuration uses available mechanisms; a new discovery service is not assumed.

Information	Purpose
Sub-network operation reference	Correlates the consumer request and its overall result.
NF and deployment requirement references	Separates required functionality from its deployment realization.
External operation and deployment references	Tracks execution and resulting managed associations.
Per-component and aggregate acceptance	Distinguishes deployment readiness, NF configuration and sub-network usability.
Partial-effect and recovery information	Identifies what exists after failure and how it will be handled.

2.2 Process-oriented procedure

- Step 1. The producer receives and authorizes the sub-network request, checks its completeness and records a stable operation reference. Conflicting or unsupported constraints are reported before execution when detectable.
- Step 2. It derives the NF functional composition and identifies which requirements can use existing NF Deployments and which require new or changed deployments. It checks that reuse is authorized and compatible with existing service dependencies.
- Step 3. It maps deployment requirements to supported external capabilities. Location or resource constraints remain requirements for the external system to evaluate; the producer does not directly schedule infrastructure resources outside its responsibility.
- Step 4. It requests the required deployment actions and records external execution references. Parallel actions are permitted when dependencies allow. The aggregate operation remains in progress while required results are outstanding.
- Step 5. For each external result, it verifies association with the current operation/desired requirement. Deployment readiness is recorded separately from NF configuration and service readiness. A late result for a superseded requirement is reconciled without automatically satisfying the current request.
- Step 6. It performs or confirms the applicable NF configuration through 3GPP management capabilities and establishes the required sub-network relationships. An external deployment success cannot substitute for missing NF configuration evidence.
- Step 7. It checks aggregate acceptance, including the required composition, relevant functional configuration and applicable connectivity/service conditions. It reports success only when the request's required conditions are satisfied.
- Step 8. On failure or cancellation, it reconciles completed and in-flight effects. It chooses authorized forward recovery, safe release of request-exclusive effects, preservation of shared resources or an explicit partial-failure result. It reports remaining effects and uncertainty rather than pretending the operation was atomic.

T12 - Safeguarded processing flow

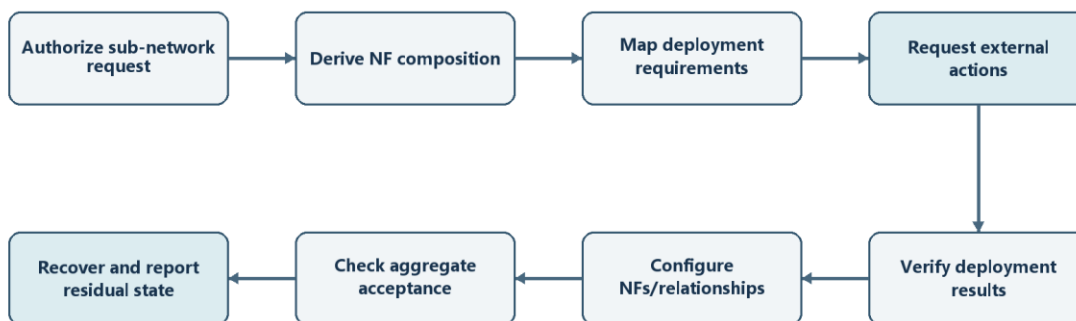


Figure T12-1. Logical interaction and safeguarded processing flow.

2.3 Declarative delegation alternative

As an alternative, the producer forms a versioned desired sub-network composition and associated NF Deployment constraints. An external management system coordinates deployment realization and reports observed state and unsatisfied constraints. The producer still performs applicable NF management and aggregate sub-network acceptance.

A changed target creates a new desired revision and an explicit relation to the earlier operation. The external system may continue work already issued for the earlier revision; its results are tracked and reconciled. Cancellation or target replacement therefore needs the same partial-effect semantics as the process-oriented option.

The two options differ in how deployment work is coordinated, not in which party decides that the original 3GPP request succeeded. They may coexist for different external domains without forcing a single GitOps pipeline or one external orchestration technology.

2.4 Failure and worked example

A requested sub-network requires two NFs. Deployment of NF-A succeeds and NF-B fails due to an unsatisfied placement constraint. The producer does not report the sub-network created. If NF-A is not shared and safe release is supported, it can request cleanup. If a later authorized request already depends on NF-A, it preserves that deployment and reports the partial state while seeking another realization for NF-B.

If NF-B's response is lost, the producer first queries the associated external operation. It does not immediately create a second NF-B deployment. If actual state cannot be established, the aggregate result remains unresolved and identifies the affected component. Changes to a previously accepted sub-network similarly preserve a clear association between old and new target versions.

2.5 Evaluation and modelling implications

Evaluation scenarios include partial deployment, configuration failure after deployment, reuse of an existing deployment, changing target composition, unsupported external constraints, response loss and cancellation. Criteria include incorrect aggregate success, duplicate deployments, damage to shared resources and transparency of residual state.

The study should compare the required association information with existing NRM before selecting new attributes or objects. It does not assume the deleted NFDeploymentParameters proposal or a fixed NF-to-deployment cardinality. Specific desired-state encodings and external scheduling algorithms remain implementation choices.

3. References and proposed agreement

Sources: S5-263622, S5-263620, S5-263624 and S5-263625, with TS 28.531 and TS 28.533 as existing provisioning/framework references. Detailed mapping must be checked against the editor baseline and the applicable current specification versions before submission.

SA5 is invited to include this candidate solution, compare process-oriented coordination with declarative delegation, and evaluate aggregate acceptance and partial-effect information as common requirements across both options.

Annex A. Submission checks and source register

Each contribution is independently extractable from its T-number heading through its proposed agreement. The T-numbers are internal drafting identifiers. “Step” labels and logical information names describe the candidate procedure and are not claims that corresponding operations or attributes already exist in 3GPP specifications.

Before registration, the submitting organization should insert its authorized Source, official meeting metadata and TDoc identifier; replace the target placement with the editor-assigned solution location; compare with the complete latest TR baseline; and remove this internal annex from the submitted individual contribution. No missing experimental result is to be filled with an assumed improvement percentage.

The local evidence baseline consists of final S5-263545, S5-263546, S5-263547 and S5-263620 through S5-263626, their meeting revision chains, and the 2026-08-28 15:30 Chairman note. The 15:33 TDoc list records final status. A merged original contribution does not establish that all its original procedure text was accepted.

Public source links for review: [Chairman note](#); [final TDoc list](#); [TR 32.801-01 portal](#).

External technical background, not mandatory realization: [RFC 9110](#); [W3C SHACL](#); [PROV-O](#); [Kubernetes API concepts](#); [Server-Side Apply](#); [Deployment API](#); [OTel service conventions](#); [OTel sampling](#).

The patent companion volume contains proposed implementation mechanisms and claim drafts. Acceptance of a TR solution does not establish patentability, ownership or standard essentiality. Publication and any applicable IPR disclosure are to be coordinated by the submitting organization.